

DATABASE MANAGEMENT SYSTEMS:

Capstone Project Handbook

MySQL Implementation Guide for Teams

(As Assigned in the Official Team Allocation Sheet)

Overview

This handbook outlines the final Capstone Project phase of the Database Management Systems (DBMS) course. Students are expected to design, implement, and present a complete relational database solution using MySQL as the primary backend technology.

Teams may optionally develop a frontend interface using any suitable technology stack (HTML/CSS/JavaScript, etc.). However, the core evaluation will focus on database design quality, normalization, SQL implementation, and backend logic.

To simulate real-world database engineering workflows, students will work in collaborative teams and implement one complete database-driven application from the project options provided below.

All projects must demonstrate proper relational schema design, entity relationships, normalization, integrity constraints, and advanced SQL operations.

Part 1: Core Capstone Projects

Each team must select ONE of the following capstone projects for implementation using MySQL as the backend database system.

The corresponding E-R Diagram for each project will be provided by the faculty. Teams are expected to analyze the supplied design and implement the complete relational database system accordingly.

Teams may further enhance or extend the provided E-R design wherever logically required.

The project must include:

- Relational Schema
 - MySQL Database Implementation
 - Sample Dataset
 - SQL Queries
 - Functional Demonstration
 - Viva Presentation
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1. Social Media Application

Focus: Complex Relationships & Interaction Systems

A database-driven social networking platform involving user interactions, posts, comments, messaging, hashtags, and group participation.

Major DBMS Concepts

- Recursive Relationships
- Many-to-Many Mapping
- Junction Tables
- Referential Integrity
- Complex Relationship Handling

[Project Link](#)

2. Grocery Management Application

Focus: E-Commerce Transactions & Inventory Management

A database system for managing customers, products, orders, suppliers, delivery systems, and payments.

Major DBMS Concepts

- Transaction Processing
- Inventory Management
- Multi-table Relationships
- Order Processing Systems
- Aggregate Query Handling

[Project Link](#)

3. Bus Reservation System

Focus: Reservation Management & Transaction Handling

A transportation reservation database system involving passengers, buses, conductors, schedules, and ticket booking workflows.

Major DBMS Concepts

- Reservation Systems
- Transaction Consistency
- Foreign Key Relationships
- Availability Management
- Booking Operations

[Project Link](#)

4. Library Management System

Focus: Classical Relational Database Design

A library database system managing books, publishers, members, and borrowing operations.

Major DBMS Concepts

- One-to-Many Relationships
- Many-to-Many Relationships
- Borrow-Return Transactions
- Data Normalization
- Relational Integrity

[Project Link](#)

Part 2: Mandatory Technical Requirements

Regardless of the project selected, every team must implement the following components.

1. Database Design

Teams must prepare:

- Relational Schema
- Data Dictionary
- Cardinality and Participation Constraints

Primary Keys and Foreign Keys must be clearly specified.

2. Normalization

The database must be normalized up to at least Third Normal Form (3NF).

Teams should demonstrate:

- Removal of redundancy
 - Elimination of anomalies
 - Functional dependency analysis
 - Proper decomposition where required
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3. MySQL Implementation

The complete backend must be implemented using MySQL.

The implementation must include:

- CREATE TABLE statements
 - ALTER operations
 - INSERT, UPDATE, DELETE operations
 - Constraints
 - Views
 - Indexes (where applicable)
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4. SQL Query Requirements

Each project must demonstrate advanced SQL usage including:

Mandatory Query Categories

- Joins
 - Nested Queries
 - Aggregate Functions
 - GROUP BY and HAVING
 - Views
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5. Dataset Requirements

Teams must populate the database with meaningful sample data.

Minimum Requirement

- At least 100+ records distributed across multiple tables.

Dummy or repetitive datasets will not be accepted.

6. Frontend Interface (Preferred)

Teams are encouraged to build a frontend application connected to the MySQL backend.

Suggested technologies:

- HTML/CSS/JavaScript

Frontend implementation is preferred but not mandatory.

Part 3: Team Collaboration Guidelines

Students are expected to work collaboratively in professional software engineering workflows.

Each member must contribute to:

- Database design
- Query writing
- Backend implementation
- Testing
- Documentation

No student may work exclusively on frontend development.

Every member must be able to explain:

- Table structures
 - Relationships
 - Constraints
 - SQL queries
 - Normalization logic
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Part 4: Unified Evaluation Criteria (Viva Presentation)

The final evaluation will focus on both technical implementation and conceptual understanding.

1. Database Design & Architecture (30%)

Evaluation Parameters

- Schema design
 - Relationship modeling
 - Normalization quality
 - Constraint implementation
 - Database structuring
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2. MySQL Implementation & Querying (35%)

Evaluation Parameters

- SQL correctness
 - Query optimization
 - Data integrity
 - Complex query execution
 - Proper implementation of constraints
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3. Functionality & System Demonstration (20%)

Evaluation Parameters

- Feature completeness
 - Data consistency
 - Error handling
 - Frontend-backend integration
 - User interaction flow
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4. Viva & Technical Defense (15%)

Evaluation Parameters

- Understanding of relationships
 - Query explanation
 - Design justification
 - Team collaboration
 - Problem-solving ability
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Part 5: Submission Requirements

Each team must submit the following:

Mandatory Deliverables

1. Project Report (PDF)
 2. Relational Schema
 3. SQL Script File (.sql)
 4. Sample Dataset
 5. Presentation Slides (PPT/PDF)
 6. Source Code (if frontend implemented)
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Important Notes

- Plagiarism will result in disqualification.
- Teams are encouraged to extend the base project creatively.

- Proper naming conventions and documentation are expected.
 - User-friendly design and modular implementation will receive additional credit.
 - Evaluation will prioritize backend database quality over UI aesthetics.
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Final Objective

The purpose of this capstone is to help students bridge academic DBMS concepts with real-world database engineering practices involving:

- Relational Modeling
- Query Design
- Transaction Management (where applicable to the selected project)
- Data Integrity
- System Architecture
- Collaborative Development

Students are expected to demonstrate not only implementation skills, but also strong conceptual understanding of database systems.